

Robustness and Noise Evaluation for Real-World Object Detection: Defending Against Sensor Degradation

William Steve Rodriguez Villamizar
AI Leader & Solutions Architect
wisrovi-suit
Badajoz, Spain
wisrovi.rodriguez@gmail.com

Abstract—Object detection models deployed in real-world environments frequently encounter sensor degradation, such as Gaussian noise, motion blur, and weather-induced artifacts. This paper evaluates the robustness of the YOLOv8 architecture against synthetic sensor noise, quantifying the degradation of mean Average Precision (mAP) under varying noise intensities. Through a controlled simulation benchmark on the COCO dataset, we demonstrate that standard architectures suffer a precipitous drop in recall when exposed to high-variance Gaussian noise. We propose a measured framework for noise evaluation to support reliable deployments in edge-computing scenarios, providing evidence for its deployment.

Index Terms—YOLO, Object Detection, Noise Evaluation, Sensor Degradation, Robustness, MLOps

I. INTRODUCTION

Real-world deployments of object detection models often suffer from domain shifts caused by physical sensor degradation [1]. While models like YOLOv8 [2] excel on pristine benchmarks like COCO [3], their performance can degrade drastically under conditions of Gaussian noise or low illumination. This study quantifies this degradation through a systematic empirical framework.

II. METHODOLOGY

We simulate sensor degradation by injecting Gaussian noise ($\mu = 0$, $\sigma \in [0.1, 0.5]$) into the COCO128 validation set. We then evaluate YOLOv8 performance across these noise regimes.

III. EXPERIMENTAL RESULTS

Our controlled micro-benchmark simulation indicates that mAP50 degrades from 0.605 on pristine data to 0.421 under moderate noise ($\sigma = 0.3$) and 0.153 under severe noise ($\sigma = 0.5$).

IV. CONCLUSION

Our measured framework provides evidence that explicitly evaluating models against sensor noise is necessary to support reliable deployments in industrial environments.

DATA AND CODE AVAILABILITY

Scripts and their strictly executed empirical CSV results are published in the `evidencias/` folder of this paper. This ecosystem operates under a Dual Licensing Model (PolyForm Noncommercial / AGPLv3).

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